Name: _____

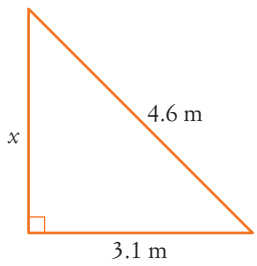
WORKSHEET

Resolution of forces

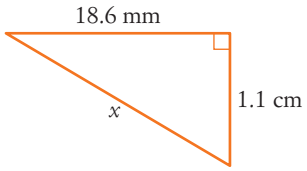
Give answers correct to two decimals answers where appropriate.

1 Find the value of x in these figures.

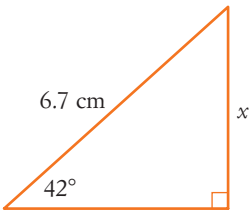
a



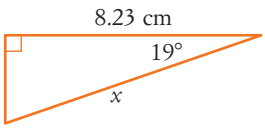
b



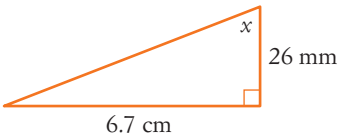
c



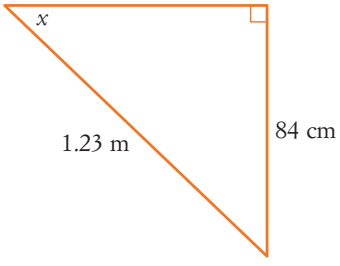
d



e



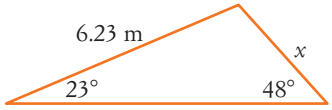
f



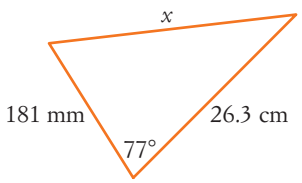
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2 Use the sine rule $\left(\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}\right)$ or the cosine rule ($c^2 = a^2 + b^2 - 2ab\cos(C)$) to find the value of x in these figures.

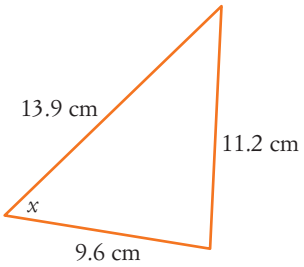
a



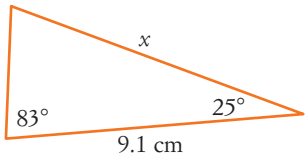
b



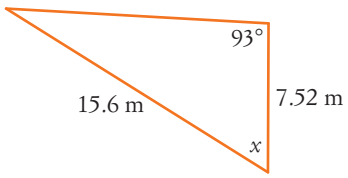
c



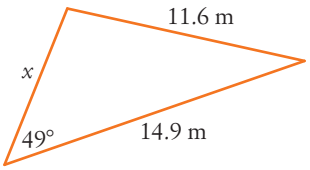
d



e



f



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3 For each of these situations:

i draw a sketch

ii resolve the forces **a** and **b** (and **c** where appropriate)

iii find the resultant vector, **r**.

iv draw a sketch showing the addition of vectors

v find the magnitude and direction of **r**.

vi state the result

a A force of 50 N is acting on an object at an angle 40° upwards from the horizontal, and another force of 30 N is acting on the same object at an angle 25° upwards from the horizontal.

b A force of 100 N is acting on an object at an angle 30° upwards from the horizontal, and another force of 70 N is acting on the same object at an angle 25° downwards from the vertical.

c An object is subjected to three forces: a force of 50 N at 35° upwards from the horizontal, 80 N at 15° upwards from the horizontal and 20 N 70° upwards from the horizontal.

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d An object is subjected to three forces: a force of 75 N at 65° upwards from the horizontal, 20 N at 40° downwards from the vertical and 100 N 70° upwards from the horizontal.

4 Find the resultant force, **r**, for the following situations.

a A force of 60 N is acting on an object at an angle 25° upwards from the horizontal, and another force of 45 N is acting on the same object at an angle 70° upwards from the horizontal.

b An object is subjected to three forces: a force of 10 N at 15° upwards from the horizontal, 60 N at 67° upwards from the horizontal and 100 N 30° downwards from the vertical.

c A force of 200 N is acting on an object at an angle 30° upwards from the horizontal, and another force of 45 N is acting on the same object at an angle 10° downwards from the horizontal.

d An object is subjected to three forces: a force of 90 N at 20° upwards from the horizontal, 150 N at 42° downwards from the horizontal and 40 N 59° downwards from the horizontal.

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5 For the following situations;

i draw a sketch showing the orthogonal components relative to the stated slope

ii find the projection **s** (parallel component)

iii find the projection **p** (perpendicular component)

iv state the resolution of this force into the components parallel and perpendicular to the slope.

a An object is being pushed up a hill with a slope of 11°. The force pushing the object is pushing at 250 N at an angle 4° downwards from the horizontal.

b An object is being pushed up a hill with a slope of 10°. The force pushing the object is pushing at 750 N at an angle 3° upwards from the horizontal.

c An object going up a hill with a slope of 6° has a weight of 850 N acting vertically downwards.

d An object going up a hill with a slope of 17° has a weight of 2500 N acting vertically downwards.

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6 Resolve the following situations' forces into parallel and perpendicular components relative to the stated slope.

a An object is being pushed up a hill with a slope of 32°. The force pushing the object is pushing at 1250 N at a horizontal angle.

b An object is being pushed up a hill with a slope of 4°. The force pushing the object is pushing at 750 N at an angle 9° upwards from the horizontal.

c An object is being pushed up a hill with a slope of 16°. The force pushing the object is pushing at 2000 N at an angle 4° downwards from the horizontal.

d An object going up a hill with a slope of 12° has a weight of 11 500 N acting vertically downwards.

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Answers

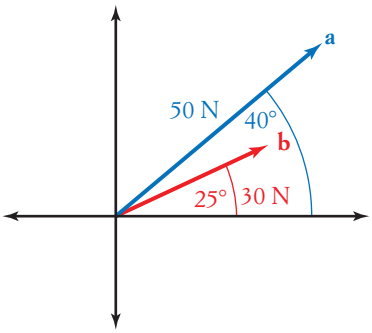
1 **a** 3.40 m **c** 4.48 cm **e** 68.79°

b 2.16 cm **d** 8.70 cm **f** 43.07°

2 **a** 6.86 m **c** 53.19° **e** 28.78°

b 28.38 cm **d** 9.50 cm **f** 6.93 m

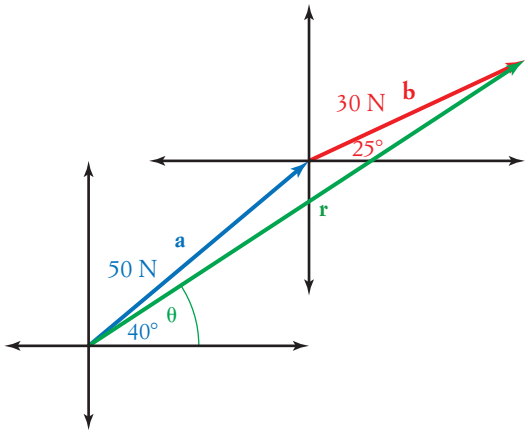
3 **a** **i**



ii $\mathbf{a} = (38.3, 32.1)$, $\mathbf{b} = (27.2, 12.7)$

iii $\mathbf{r} = (65.5, 44.8)$

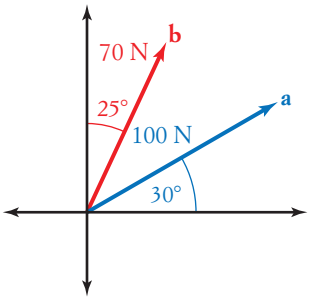
iv



v $|\mathbf{r}| = 79.4$ and $\theta = 34.4°$.

vi The resultant force is approximately 79.4 newtons at 34.4° to the horizontal.

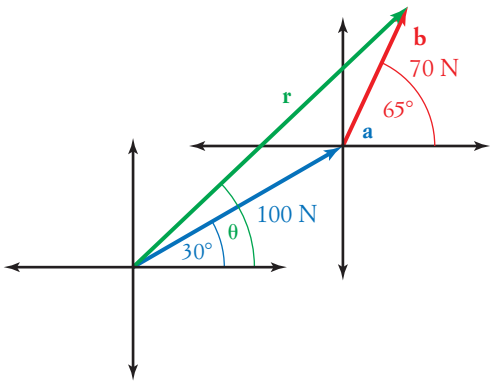
b **i**



ii $\mathbf{a} = (86.6, 50.0)$, $\mathbf{b} = (29.6, 63.4)$

iii $\mathbf{r} = (116.2, 113.4)$

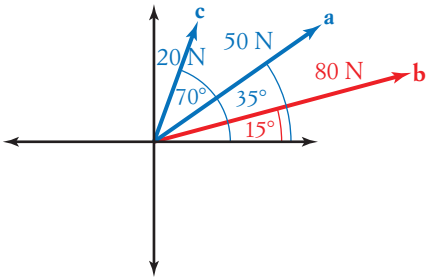
iv



v $|\mathbf{r}| = 162.4$ and $\theta = 44.3°$.

vi The resultant force is approximately 162.4 newtons at 44.3° to the horizontal.

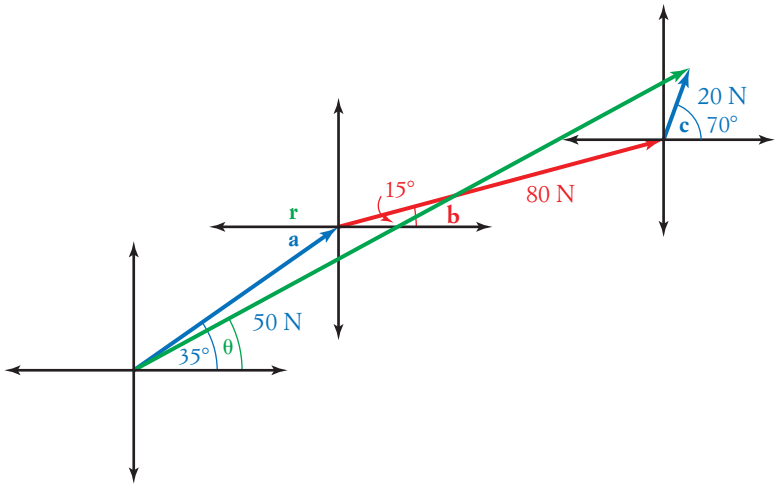
c i



ii $\mathbf{a} = (41.0, 28.7)$, $\mathbf{b} = (77.3, 20.7)$, $\mathbf{c} = (6.8, 18.8)$

iii $\mathbf{r} = (125.1, 68.2)$

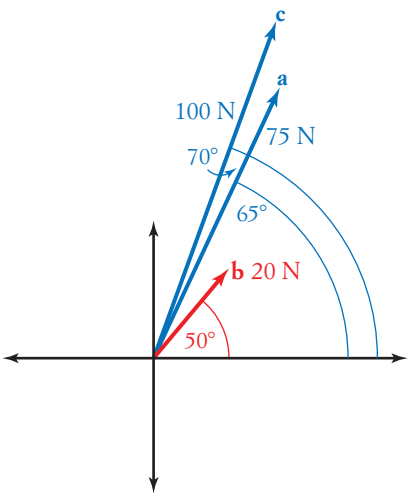
iv



v $|\mathbf{r}| = 142.4$ and $\theta = 28.6°$.

vi The resultant force is approximately 142.4 newtons at 28.6° to the horizontal.

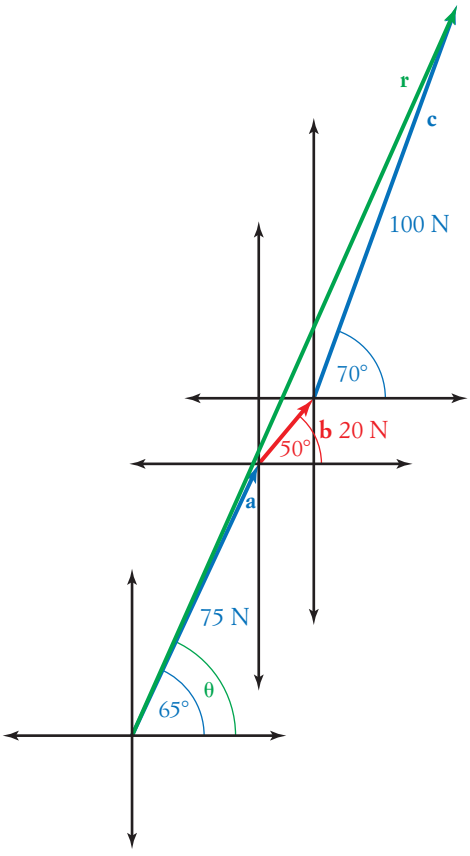
d i



ii $\mathbf{a} = (31.7, 68.0)$, $\mathbf{b} = (12.9, 15.3)$, $\mathbf{c} = (34.2, 94.0)$

iii $\mathbf{r} = (78.8, 177.3)$

iv



v $|\mathbf{r}| = 194.0$ and $\theta = 66.0°$.

vi The resultant force is approximately 194.0 newtons at 66.0° to the horizontal.

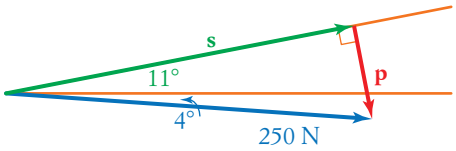
4 a The resultant force is approximately 97.2 newtons at 44.1° to the horizontal.

b The resultant force is approximately 166.6 newtons at 60.1° to the horizontal.

c The resultant force is approximately 249.0 newtons at 21.1° to the horizontal.

d The resultant force is approximately 240.3 newtons at 25.6° downwards from the horizontal.

5 a i

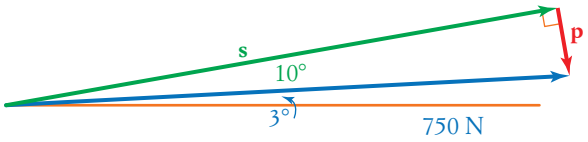


ii $\mathbf{s} = 250 \cos 15° = 241.5$

iii $\mathbf{p} = 250 \cos 75° = 64.7$

iv The components of the 250 N force parallel and perpendicular to the slope are about 241.5 N and 64.7 N respectively.

b i

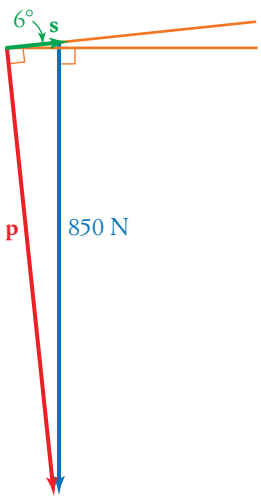


ii $\mathbf{s} = 270 \cos 7° = 744.4$

iii $\mathbf{p} = 750 \cos 83° = 91.4$

iv The components of the 750 N force parallel and perpendicular to the slope are about 744.4 N and 91.4 N respectively.

c i

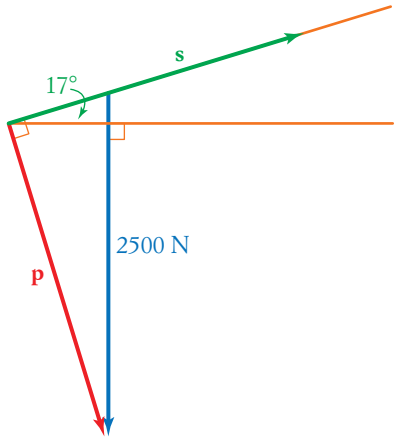


ii $\mathbf{s} = 850 \cos 84° = 88.8$

iii $\mathbf{p} = 850 \cos 6° = 845.3$

iv The components of the 850 N force parallel and perpendicular to the slope are about 88.8 N and 845.3 N respectively.

d i



ii $\mathbf{s} = 2500 \cos 73° = 730.9$

iii $\mathbf{p} = 2500 \cos 17° = 2390.8$

iv The components of the 850 N force parallel and perpendicular to the slope are about 730.9 N and 2390.8 N respectively.

6 a The components of the 1250 N force parallel and perpendicular to the slope are about 1060.1 N and 662.4 N respectively.

b The components of the 750 N force parallel and perpendicular to the slope are about 747.1 N and 65.4 N respectively.

c The components of the 2000 N force parallel and perpendicular to the slope are about 1879.4 N and 684.0 N respectively.

d The components of the 11500 N force parallel and perpendicular to the slope are about 2391.0 N and 11248.7 N respectively.

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